Significance test

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ttest

```
roll <- c(23,45,66,34,54)
marks <- c(87,89,80,98,87)
student_data <- data.frame(roll,marks)
student_data$group <- ifelse(student_data$roll < 50, "Group1", "Group2")
table(student_data$group)
```

```
Group1 Group2 
     3      2 
```

```
ttest <- t.test(marks ~ group,data = student_data)
ttest
```

```

	Welch Two Sample t-test

data:  marks by group
t = 1.6092, df = 2.6046, p-value = 0.2194
alternative hypothesis: true difference in means between group Group1 and group Group2 is no
95 percent confidence interval:
```

```
 -9.076982 24.743649
sample estimates:
mean in group Group1 mean in group Group2
            91.33333             83.50000
```

One Way ANOVA

```r
Stu_data <- data.frame(
Method = rep(c("A", "B", "C"), each = 5),
Score = c(85, 90, 88, 87, 84,  # Method A
78, 76, 80, 75, 79,  # Method B
92, 94, 95, 91, 93)  # Method C
)

anova_result <- aov(Score ~ Method, data = Stu_data)
summary(anova_result)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)    
Method       2  600.4  300.20   72.05 2.06e-07 ***
Residuals   12   50.0    4.17                     
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- aov: A function in R for conducting ANOVA.
- Score: The dependent variable or the response variable.
- Method: The independent variable or the factor whose levels you are

Look at the p-value in the output:

1. If p-value < 0.05, reject the null hypothesis (there is a significant difference between the groups).
2. If p-value 0.05, fail to reject the null hypothesis (there is no significant difference).

The null hypothesis (H) is the default assumption that there is no difference between the groups.